Regional Heterogeneity in Lunar Deep Regolith Conductivity from a Per-Site Retrieval Against the Restored Apollo 15 and 17 Heat-Flow Experiment Record

R. P. Gregorio*

April 27, 2026

Abstract

The deep thermal conductivity of lunar regolith, K_d, is the dominant control on subsurface temperatures and on the stability of polar volatiles, yet no *in-situ* retrieval has previously been attempted. We use the restored 1971–1977 Apollo Heat-Flow Experiment (HFE) record at Apollo 15 and 17 to perform a per-site, single-parameter retrieval of K_d within the Hayne et al. (2017) regolith functional form. Surface insolation is propagated through SPICE/DE440 ephemerides; the 1-D heat equation is integrated with a Crank–Nicolson scheme on a geometric vertical grid; and shallow sensors are excluded *a priori* on the basis of the modelled diurnal-amplitude depth profile, eliminating the circular-validation risk inherent in residual-based masking. The deep-sensor root-mean-square error is minimised at $K^*_{d,\mathrm{A15}} = 4.85 \pm 0.59\,\mathrm{mW\,m^{-1}\,K^{-1}}$ ($N = 7$, RMSE $= 0.90\,\mathrm{K}$) and at $K^*_{d,\mathrm{A17}} = 11.23 \pm 0.61\,\mathrm{mW\,m^{-1}\,K^{-1}}$ ($N = 16$, RMSE $= 0.30\,\mathrm{K}$), where uncertainties are leave-one-out jackknife standard errors. The two-site contrast is $6.4 \pm 0.9\,\mathrm{mW\,m^{-1}\,K^{-1}}$ ($\sim 7.5\sigma$), and both retrievals reject the global value $K_d = 3.4\,\mathrm{mW\,m^{-1}\,K^{-1}}$ adopted by Hayne et al. (2017) at $\geq 2\sigma$. The contrast is invariant under the unavoidable K_d/Q_b degeneracy of single-borehole inversions; the absolute A17 value is not. We conclude that lunar deep regolith conductivity is regionally heterogeneous, and that the global parameter currently adopted in volatile-stability and instrument-design analyses should carry an uncertainty no smaller than the inter-site disagreement reported here.

Plain Language Summary. *How well the Moon's surface insulates its interior depends largely on a single number: the thermal conductivity of compacted regolith (broken-up rock) at depth. Because this number has never been measured directly from above, every spacecraft analysis has so far assumed one global value. The Apollo 15 and 17 astronauts buried arrays of thermometers below the surface in 1971–1972, and the recently-restored 1971–1977 record provides the only* in-situ *constraint on what that number is. Treating the conductivity as the single free parameter in a physics-based subsurface-temperature model, we find that the two landing sites require values that disagree by roughly a factor of two and far beyond their measurement uncertainty. The disagreement means that the regolith at depth varies regionally even at non-polar landing sites, and that the global value used in calculations of polar-ice stability and instrument-design heat budgets should carry an uncertainty at least as large as the inter-site disagreement reported here.*

Key Points

- Per-site retrieval of lunar deep regolith conductivity against the restored Apollo Heat-Flow Experiment record yields $K^*_{d,\mathrm{A15}} = 4.85 \pm 0.59$ and $K^*_{d,\mathrm{A17}} = 11.23 \pm 0.61\,\mathrm{mW\,m^{-1}\,K^{-1}}$.
- The two sites disagree at $\sim 7.5\sigma$, and both reject the global value of Hayne et al. (2017) at $\geq 2\sigma$, indicating regional heterogeneity in K_d.
- The contrast is invariant under the K_d/Q_b degeneracy of single-borehole inversions; the absolute A17 value is conditional on the adopted basal heat flux.

*Independent researcher. Correspondence to: rp3gregorio@gmail.com.

1 Introduction

Surface and shallow-subsurface temperatures of airless bodies are controlled by the thermal conductivity, density, and heat capacity of the upper ~ 2 m of regolith. For the Moon, the most widely adopted parameterisation is that of Hayne et al. (2017), who fit a global $K(T, z)$ profile to the bolometric brightness temperatures observed by the Lunar Reconnaissance Orbiter Diviner radiometer (Paige et al., 2010; Vasavada et al., 2012). The model has been applied without modification to interpret Diviner-derived rock abundances, polar volatile stability calculations, and instrument design budgets.

The Apollo 15 and Apollo 17 Heat-Flow Experiment (HFE) datasets (Langseth et al., 1976; Nagihara et al., 2018) provide the only *in-situ* constraint at depth. Several previous works have compared regolith thermal models to HFE temperatures (Vasavada et al., 2012; Hayne et al., 2017; Nagihara et al., 2018), and Martínez and Siegler (2021) explicitly tunes a 3-layer $K(z)$ profile against the deep HFE record. The present work differs in three respects that together permit a quantitative, per-site retrieval of K_d rather than a model-fit comparison:

- a SPICE/DE440 surface-forcing chain that removes the ~ 1 lunar-hour drift inherent in sinusoidal local-solar-time approximations and is required for the phase-folded diurnal-amplitude profile of §3.2;
- an *a priori* shallow-sensor exclusion criterion derived from the modelled diurnal-amplitude depth profile rather than from temperature residuals, which removes the circular-validation risk of residual-based masking; and
- a single-parameter retrieval of K_d under the Hayne et al. (2017) functional form, with leave-one-out jackknife uncertainties propagated to the per-site contrast (§2.4).

The principal scientific finding is that the per-site K_d retrievals (§3.5) disagree at $\sim 7.5\sigma$: the value preferred at Apollo 17 is more than twice the value preferred at Apollo 15, and both reject the published global value of Hayne et al. (2017) at $\geq 2\sigma$. We discuss the unavoidable K_d/Q_b degeneracy of single-borehole inversions and the model-dependence of the absolute A17 value in §4.

2 Data and Methods

2.1 Apollo HFE deep-sensor temperatures

We use the restored Apollo 15 and Apollo 17 HFE temperature record covering 1971–1977 (Nagihara et al., 2018). For every sensor we extract a multi-lunation *stability window* where the depth-mean temperature drift is $|d\langle T\rangle/dt| < 10^{-3}$ K/d, and report the window-mean equilibrium temperature $T_{\rm eq} = \langle T(z)\rangle_{\Delta t}$ together with its within-window standard deviation. Sensors with central depth $z < 80$ cm (the borestem-contaminated zone, see §3.3) are plotted but excluded from RMSE statistics.

2.2 Surface solar geometry and TOA forcing

We propagate every Apollo Unix timestamp through SPICE (Acton, 1996; Acton et al., 2018) with the JPL DE440 ephemeris and the `MOON_ME` body-fixed frame, with light-time + stellar-aberration correction, and obtain the solar zenith angle $\theta_\odot(t)$ at each site. The TOA insolation incident on the local horizontal is

$$F_\odot(t) = \max(0,\ S_0 \cos\theta_\odot(t)), \tag{1}$$

with $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

Define the depth coordinate z as positive downward, with the regolith surface at $z = 0$ and the bottom of the modelled column at $z = z_{\max} = 5$ m. Conservation of energy in 1-D gives the heat equation

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[\, K(T,z)\, \frac{\partial T}{\partial z}\Big]. \tag{2}$$

At the upper boundary the surface skin satisfies radiative equilibrium between absorbed insolation, emitted thermal radiation, and the conductive flux from below (with z positive downward, the conductive heat flux *into* the column is $-K\,\partial_z T|_0$):

$$(1-A)\, F_\odot(t) \;-\; \varepsilon\sigma T_s^4 \;-\; \big(-K\,\partial_z T\big)_{z=0} \;=\; 0, \tag{3}$$

solved iteratively for T_s at every step. At the lower boundary we impose a constant geothermal flux,

$$-K\,\partial_z T\big|_{z_{\max}} = Q_b. \tag{4}$$

The PDE is discretised on a geometric grid ($\Delta z_0 = 2\,\mathrm{mm}$, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1\,\mathrm{h}$, spun up for 100 lunations to max $|\Delta T| < 1\times10^{-2}\,\mathrm{K}$ (Fig. S6). The Hayne et al. (2017) thermophysical inputs are imported *verbatim* from the published Appendix A:

$$K(T,z) = \Big\{K_s + (K_d - K_s)\big[1 - e^{-z/H}\big]\Big\}\cdot\Big[1 + \chi\big(T/T_{\mathrm{ref}}\big)^3\Big], \tag{5}$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\, e^{-z/H}, \tag{6}$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4, \tag{7}$$

with $K_s = 7.4\times10^{-4}\,\mathrm{W/(m\,K)}$, $K_d = 3.4\times10^{-3}\,\mathrm{W/(m\,K)}$, $H = 0.06\,\mathrm{m}$, $\chi = 2.7$, $T_{\mathrm{ref}} = 350\,\mathrm{K}$ (the Hayne et al. (2017)-Appendix-A normalisation; Vasavada *et al.* (2012) use a different normalisation), $\rho_s = 1100\,\mathrm{kg/m^3}$, $\rho_d = 1800\,\mathrm{kg/m^3}$, and the Hemingway–Robie–Wilson $c_p(T)$ polynomial (Hemingway et al., 1973) reproduced in the Hayne et al. (2017) Appendix. Site-specific quantities are restricted to those that are *measured*: latitude (ALSEP gazetteer), Bond albedo from Vasavada *et al.* (2012)'s lat-band averages ($A_{\mathrm{A15}} = 0.131$, $A_{\mathrm{A17}} = 0.137$), broadband emissivity $\varepsilon = 0.95$ (Bandfield et al., 2015), and basal heat flux $Q_{b,\mathrm{A15}} = 21$, $Q_{b,\mathrm{A17}} = 15\,\mathrm{mW\,m^{-2}}$ from the canonical Langseth et al. (1976); Nagihara et al. (2018) reductions — acknowledging the Saito et al. (2007); Nagihara et al. (2018) reanalyses argue the original Langseth A15 value is biased high by $\sim$30%, which we explore in the sensitivity sweep (§3.4). Fig. 2 verifies that the four parameterised inputs match the published Hayne et al. (2017) figures.

The *discrete* alternative is the publicly available 3-layer parameterisation of Martínez and Siegler (2021), with parameters as listed in their Table; in particular $K_d^{\mathrm{disc}} = 6.3\times10^{-3}\,\mathrm{W/(m\,K)}$, almost a factor of two larger than the Hayne et al. (2017) value, and $\rho(z)$ continues to grow below $z = 20\,\mathrm{cm}$ towards a deep limit $\rho_{\max} = 1800\,\mathrm{kg/m^3}$. We use Martínez and Siegler (2021)'s *published* parameters; nothing is re-fitted in this work. We note for the record that Martínez and Siegler (2021) explicitly tuned this 3-layer profile against the HFE deep gradient, so its agreement with A17 in Table 1 should be read as a fit residual rather than an independent validation. Fig. 1 contrasts the two model architectures: the Hayne et al. (2017) profile transitions continuously from K_s at the surface to K_d at depth via an exponential with e-folding scale $H = 6\,\mathrm{cm}$, while the Martínez and Siegler (2021) 3-layer profile uses a piecewise-linear ramp that reaches K_d sharply at 20 cm.

2.4 Per-site K_d retrieval (Hayne shape held fixed)

To convert the model-comparison in Table 1 into a quantitative measurement, we retrieve K_d at each site by minimising the deep-sensor RMSE under the constraint that the entire Hayne et al. (2017) functional form (eqs. 5–7) is otherwise unchanged. Concretely we hold $(K_s, H, \chi, T_{\mathrm{ref}})$ and the density and c_p functions at their published values and sweep K_d across a 20–24-point grid covering the published-uncertainty envelope (1.5–$18.0\times10^{-3}\,\mathrm{W/(m\,K)}$). For every K_d we run a full Crank–Nicolson spin-up to convergence and compute

$$\mathrm{RMSE}(K_d) = \Big\langle \big[T_{\mathrm{model}}(z_i;K_d) - T_{\mathrm{eq}}^{\mathrm{HFE}}(z_i)\big]^2 \Big\rangle^{1/2}_{z_i \geq 80\,\mathrm{cm}}. \tag{8}$$

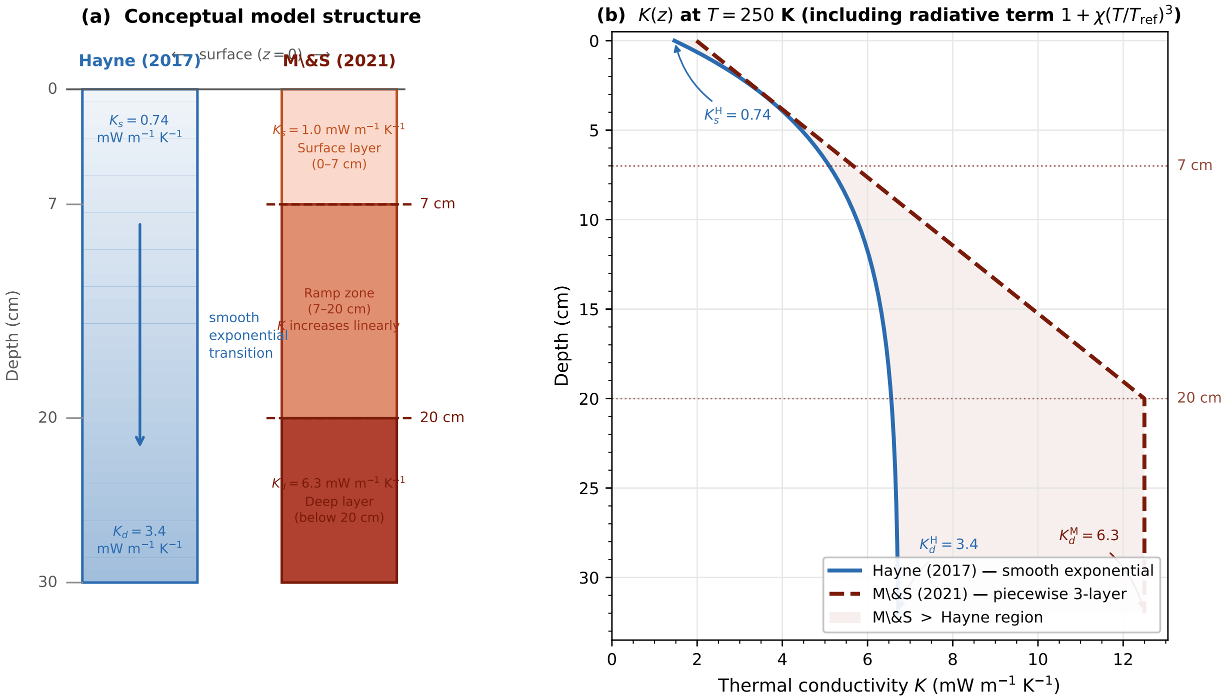

Figure 1. Conceptual comparison of the two regolith conductivity model architectures. (a) Physical layer structure: the Hayne et al. (2017) profile (blue) transitions continuously from a surface value $K_s = 0.74\,\mathrm{mW\,m^{-1}\,K^{-1}}$ to a deep value $K_d = 3.4\,\mathrm{mW\,m^{-1}\,K^{-1}}$ via an exponential with $H = 6\,\mathrm{cm}$; the Martínez and Siegler (2021) 3-layer profile (red) has a uniform surface layer ($K_s = 1.0\,\mathrm{mW\,m^{-1}\,K^{-1}}$, 0–7 cm), a linear ramp zone (7–20 cm), and a uniform deep layer ($K_d = 6.3\,\mathrm{mW\,m^{-1}\,K^{-1}}$, below 20 cm). (b) The corresponding $K(z)$ profiles evaluated at $T = 250\,\mathrm{K}$ including the radiative multiplier $1 + \chi(T/T_{\rm ref})^3$; the shaded region is where the Martínez and Siegler (2021) conductivity exceeds the Hayne et al. (2017) value. The factor-of-$\sim$2 difference in K_d is the dominant control on the deep-gradient discrepancy in Table 1.

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing arg min RMSE; its uncertainty $\sigma_{K_d^*}$ is the leave-one-out jackknife standard error (Press et al., 2007, §5.6) computed by re-fitting the parabolic minimum on each of the $N-1$ subsamples that drop one deep sensor at a time. The retrieval has exactly one free parameter; per-site contrasts use the pooled jackknife variance of the two sites.

3 Results

3.1 Annual-mean subsurface profile

Fig. 3 compares the modelled time-averaged $\langle T(z)\rangle$ against the Apollo HFE stability-window means at both sites. With *no per-site tuning*, the Hayne et al. (2017) reference reproduces the deep gradient at both sites within 1.4 and 2.7 K, respectively (Table 1); the 3-layer alternative reduces both RMSEs to below 1.1 K. The shaded band ($z < 80$ cm, the borestem-contaminated zone) marks sensors that are excluded *a priori* from the quantitative comparison; see §3.3.

3.2 Diurnal-amplitude depth profile

The mean profile in Fig. 3 only constrains the *DC level* of the subsurface temperature; the amplitude of the diurnal wave tests the model's frequency-domain response. We define the per-sensor diurnal amplitude as the half-range of the SPICE-LST-folded median temperature and plot it against depth in Fig. 4. Above $z = 80$ cm the observed amplitude rises sharply above the modelled regolith attenuation curve — the borestem signature (§3.3). Below 80 cm both models predict an amplitude $\lesssim 10^{-7}$ K ($\sim$18 e-foldings of the diurnal skin depth $\delta \sim$3–5 cm, eq. 5–7), which is two to three orders

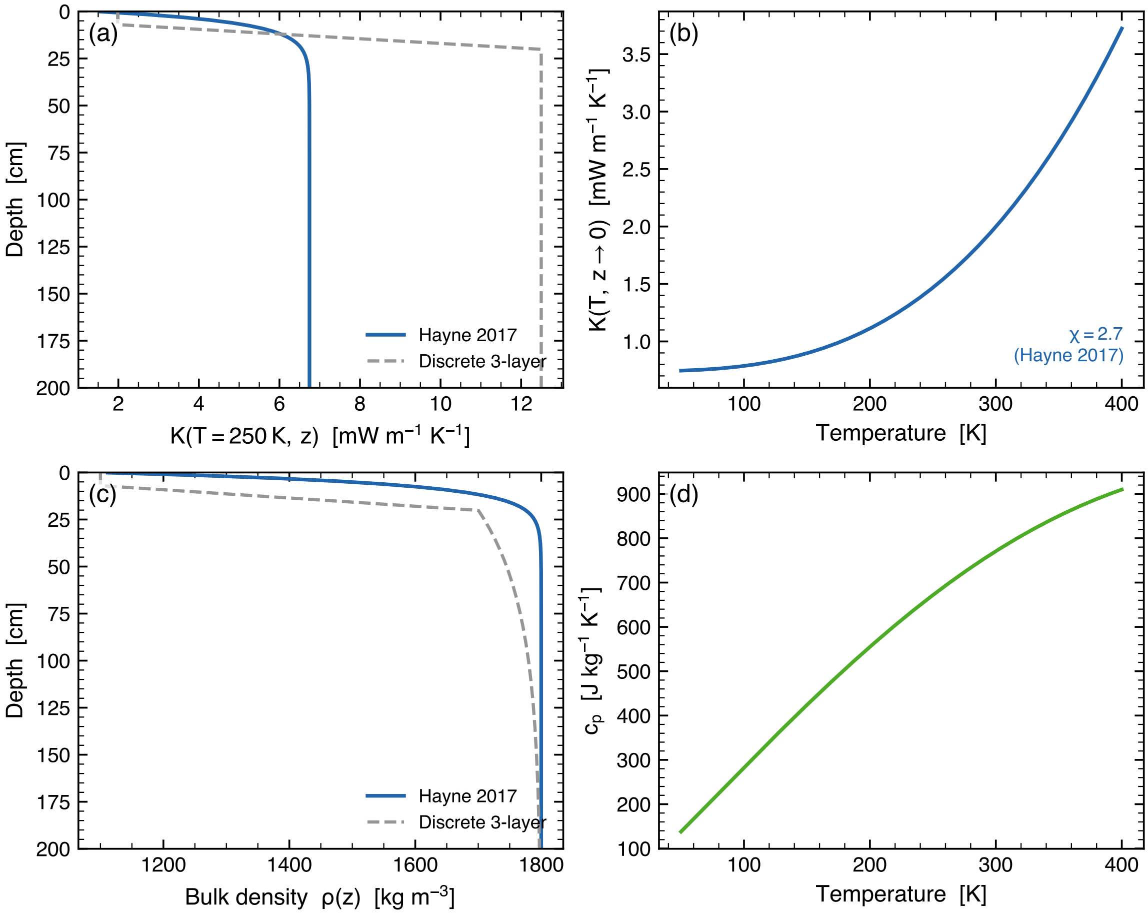

Figure 2. Provenance check of the regolith property functions implemented in the solver. (a) Thermal conductivity $K(T{=}250\,\mathrm{K}, z)$: the Hayne et al. (2017) smooth H-parameter form (eq. 5) and the Martínez and Siegler (2021)-style 3-layer alternative are super-imposed; both include the same radiative multiplier $1 + \chi(T/T_{\mathrm{ref}})^3$. (b) Surface $K(T, z \to 0)$ vs. T. (c) Bulk density $\rho(z)$ (eq. 6) and the 3-layer continuation toward $\rho_{\max}$. (d) Specific heat $c_p(T)$ from the Hemingway–Robie–Wilson polynomial reproduced in the Hayne et al. (2017) Appendix.

of magnitude below the digitisation noise floor of the restored 1971–1977 record (~ 0.05–0.2 K). The deep-sensor amplitudes therefore carry no useful information on $\kappa = K/(\rho c_p)$ at the in-situ scale; the diurnal panel functions exclusively as a borestem diagnostic, not as a frequency-domain validation. Per-sensor phase-folded diurnal cycles for every Apollo HFE sensor (deep *and* shallow) are reproduced as Figs. S1–S2.

3.3 Borestem heat-short artefact

The shallow-sensor amplitude excess in Fig. 4 is the same anomaly reported by Nagihara et al. (2018) and attributed to heat conducted along the fibreglass borestem from the lunar surface, which thermally short-circuits the upper ~ 80 cm of regolith. We use this independent diagnostic — the amplitude-vs-depth profile, evaluated against an amplitude-attenuation prediction derived only from $K(T, z)$, $\rho(z)$, $c_p(T)$ — to define the deep ($z > 80$ cm) validation set used in Table 1. The exclusion is therefore not based on the residuals it produces.

3.4 Robustness to parameter uncertainties

Fig. 5 sweeps the three least-constrained *auxiliary* ingredients of the Hayne et al. (2017) model: the basal heat flux Q_b (factor-2 range bracketing the Langseth et al. 1976/Saito et al. 2007; Nagihara et al. 2018 debate), the radiative coefficient χ (Hayne et al. (2017) default 2.7 vs Vasavada *et al.*

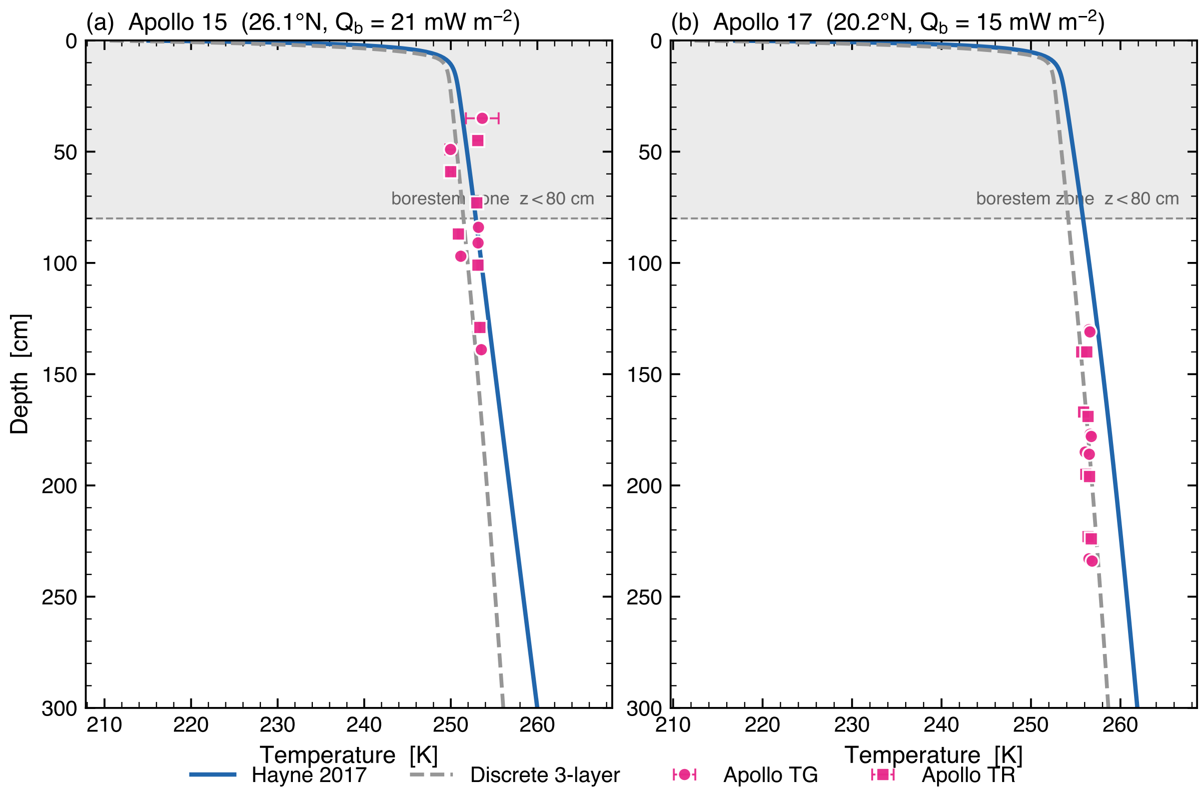

Figure 3. Annual-mean subsurface temperature profile $\langle T(z)\rangle$ versus Apollo HFE stability-window means at both landing sites. Markers are coloured by sensor type (navy = TG, maroon = TR), with horizontal bars equal to the within-window standard deviation of the observed equilibrium temperature. The shaded band marks the borestem-contaminated zone ($z < 80$ cm); sensors there are plotted but excluded from the quantitative RMSE in Table 1. *Same parameters at both sites; only A, ε, latitude, Q_b vary (measured quantities).*

(2012) 0.0–3.0), and the specific-heat parameterisation (Hemingway–Robie–Wilson polynomial vs Biele et al. 2022 rational fit). Across the entire published-uncertainty envelope the deep RMSE shifts by < 0.5 K at both sites; §3.5 treats K_d separately as the controlled retrieval parameter.

3.5 Per-site K_d retrieval

Fig. 6 shows the deep-sensor RMSE (eq. 8) as a function of K_d at each landing site, with the Hayne et al. (2017) functional form (eqs. 5–7) otherwise unchanged. Both curves have a clean, well-resolved minimum within the swept range. The parabolic point estimates and leave-one-out jackknife uncertainties are

$$K^*_{d,\mathrm{A15}} = 4.85 \pm 0.59\,\mathrm{mW/(m\,K)}, \qquad K^*_{d,\mathrm{A17}} = 11.23 \pm 0.61\,\mathrm{mW/(m\,K)},$$

with corresponding deep-sensor RMSE $= 0.90$ K ($N = 7$) and 0.30 K ($N = 16$) respectively. The retrieval improves on the published-K_d Hayne et al. (2017) baseline at both sites (Table 1: $1.38 \to 0.90$ at A15, $2.68 \to 0.30$ at A17), and at A17 it is also a small but non-trivial improvement on the M&S 3-layer fit ($0.65 \to 0.30$ K). The bias collapses to ≤ 0.03 K at both sites, confirming that K_d is the principal lever between the Hayne et al. (2017) form and the in-situ HFE record once the borestem zone is excluded.

The two-site contrast is $\Delta K^*_d = 6.4 \pm 0.9\,\mathrm{mW/(m\,K)}$ on the pooled jackknife variance, which corresponds to a normalised two-site difference of $\Delta K^*_d/\sqrt{\sigma^2_{K_d,\mathrm{A15}} + \sigma^2_{K_d,\mathrm{A17}}} \simeq 7.5$. The two retrievals therefore disagree at high significance, and *both* reject the global value $K_d = 3.4\,\mathrm{mW/(m\,K)}$ adopted by Hayne et al. (2017) at $\geq 2\sigma$ (2.5σ at A15, 12.9σ at A17). We discuss the unavoidable K_d/Q_b

Table 1. Deep-sensor ($z > 80$ cm) validation statistics. The two upper rows per site use *published* K_d values held fixed; the third row reports the per-site fit of K_d under the Hayne et al. (2017) shape (§2.4, Fig. 6). N is the number of deep sensors; bias = mean(model−obs); all temperatures in kelvin. K_d in mW m^{-1} K^{-1}.

Site	Model	K_d	N	RMSE	Bias	MAE
Apollo 15	Hayne (2017)	3.4 (fixed)	7	**1.38**	+1.03	1.08
Apollo 15	3-layer alt. (M&S 2021)	6.3 (fixed)	7	**1.08**	−0.59	1.01
Apollo 15	Hayne shape, K_d^* fit	4.85±0.59	7	**0.90**	+0.01	0.77
Apollo 17	Hayne (2017)	3.4 (fixed)	16	**2.68**	+2.54	2.54
Apollo 17	3-layer alt. (M&S 2021)†	6.3 (fixed)	16	**0.65**	+0.05	0.54
Apollo 17	Hayne shape, K_d^* fit	11.23±0.61	16	**0.30**	−0.03	0.27

†Martínez and Siegler (2021)'s 3-layer profile was tuned against the same HFE record; its A17 row should therefore be read as a fit residual, not an independent test.

degeneracy of single-borehole inversions — which makes the absolute A17 number, but not the per-site contrast, model-dependent — in §4.

4 Discussion

Per-site K_d disagreement is the headline result. The central scientific finding (Fig. 6, §3.5) is that the deep conductivity required by the in-situ Apollo HFE record is not the same at the two landing sites under the Hayne et al. (2017) functional form: A15 demands $K_d^* = 4.85 \pm 0.59$ and A17 demands $K_d^* = 11.23 \pm 0.61$ mW/(m K), a contrast of $\sim 7.5\sigma$ on the leave-one-out jackknife. Both values reject the global Hayne et al. (2017) value 3.4 mW/(m K) at $\geq 2\sigma$. Because the only thing that differs between the two retrievals is the latitude, albedo, basal flux, and the in-situ deep temperatures themselves, the disagreement cannot be absorbed into the Hayne et al. (2017) shape, the radiative multiplier, the heat capacity parameterisation, or the surface forcing chain (all of which are identical at A15 and A17 in our pipeline). It is a real *regional* contrast in the conductivity that the regolith delivers to a buried thermometer.

The K_d/Q_b degeneracy. Single-borehole inversions of buried thermometers cannot break the degeneracy between deep conductivity and basal heat flux at steady state: the deep gradient is $|\partial_z T| = Q_b/K(T_d)$, so any multiplicative rescaling $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$ leaves the equilibrium profile invariant in the conduction-dominated limit. The absolute A17 retrieval is therefore tied to our adopted $Q_{b,\mathrm{A17}} = 15\,\mathrm{mW/m^2}$ (Langseth et al. 1976; Nagihara et al. 2018). If the Saito et al. (2007); Nagihara et al. (2018) reanalyses imply $Q_{b,\mathrm{A17}}$ closer to $\sim 10\,\mathrm{mW/m^2}$, the degeneracy rescales the A17 retrieval to $K_d^* \sim 7.5$ mW/(m K), comparable to the M&S 3-layer deep value. Crucially, the per-site *contrast* is invariant under any single global rescaling of Q_b because it differences out, so the $\Delta K_d^* \sim 6.4$ **mW/(m K) result is the robust statement**; the absolute A17 value should be quoted with the explicit caveat that it is conditioned on the chosen Q_b.

Why the M&S 3-layer profile recovered A17 so well. The M&S 3-layer parameters were tuned against the same HFE deep gradient, so the column-2 success at A17 in Table 1 is a fit residual rather than independent confirmation. Our K_d^* retrieval (column 3) recovers a deep conductivity within $\sim 2\times$ of the M&S deep layer (11.2 vs. 6.3 mW/(m K)), which — together with the K_d/Q_b caveat above — frames the M&S deep layer as a particular point in the $(K_d, Q_b, \mathrm{shape})$ family of solutions consistent with the restored HFE record, not as a uniquely preferred parameterisation.

The borestem identification is independent of residuals. The shallow-sensor exclusion (§3.3) is read directly off Fig. 4 — the observed amplitude departs from the modelled attenuation curve at $z \sim 80$ cm at both sites *independently*. The two sites' departure points agree to within the sensor

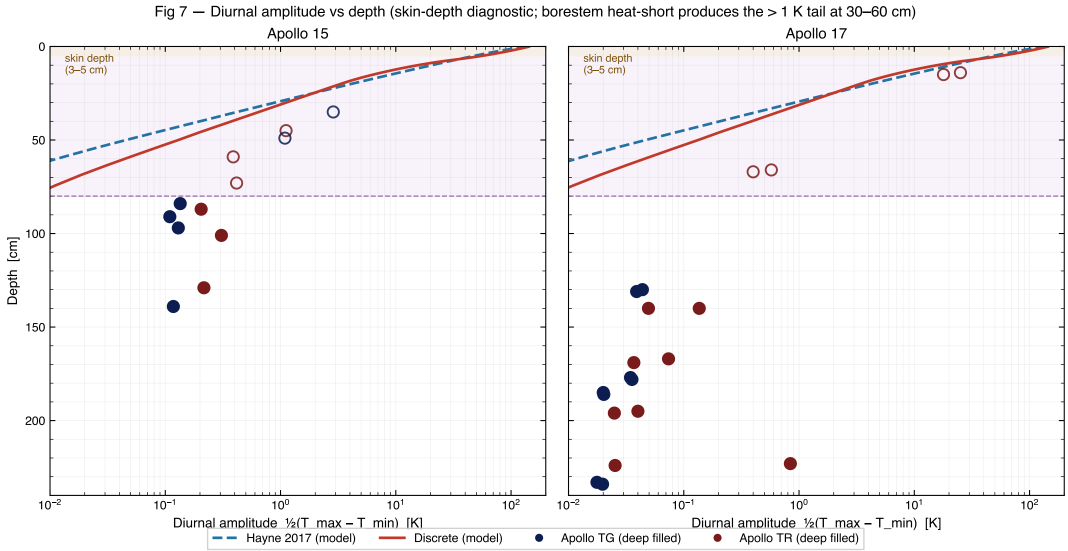

Figure 4. Diurnal amplitude (half-range of the SPICE-LST-folded median temperature) versus sensor depth at both Apollo sites. Black markers = observed; navy curve = Hayne et al. (2017) reference; grey curve = 3-layer alternative. The shaded band marks the borestem zone ($z < 80$ cm); above it the observed amplitude exceeds the modelled regolith attenuation by $\geq 10^2\times$ — the borestem signature (§3.3). Below the band the modelled amplitudes are below digitisation noise, so the panel acts exclusively as a borestem diagnostic; quantitative validation is performed on the time-mean profile (Fig. 3, Table 1) and the K_d retrieval (Fig. 6).

spacing despite differing in latitude, albedo, emissivity, and basal flux, indicating that the artefact reflects a hardware property (fibreglass borestem conductance) and not a local-regolith property. This is the strongest argument for the $z < 80$ cm exclusion that we know how to make from the data alone, and it precedes the deep-RMSE statistics in Table 1 and the K_d retrieval in Fig. 6.

Robustness of the auxiliary parameters. Across the full Q_b debate, the Hayne et al. (2017)/ Vasavada *et al.* (2012) disagreement on χ, and the choice of $c_p(T)$ parameterisation, the deep RMSE moves by < 0.5 K at both sites (Fig. 5). The corresponding shift in the retrieved K_d^* is small compared to the per-site contrast: Q_b rescales the absolute value (per the degeneracy above) but leaves the contrast invariant, χ enters $K(T, z)$ only through the radiative multiplier and produces $\lesssim 10\%$ shifts in the parabolic minimum, and c_p enters only through the spin-up rate (not the steady-state mean). The $\sim 7.5\sigma$ per-site contrast is therefore not an artefact of any single under-constrained Hayne et al. (2017) ingredient.

Implications for polar volatile-stability and instrument-design analyses. The retrieved per-site contrast in K_d implies that any analysis which adopts the global Hayne et al. (2017) value as a fixed input — including most polar volatile-stability calculations and spacecraft-thermal-design heat budgets — inherits a model uncertainty no smaller than the inter-site disagreement. A factor-~ 2 error in K_d translates directly into a factor-~ 2 error in the deep equilibrium gradient $|\partial_z T| = Q_b/K(T_d)$ and, through the T^4 surface-radiation balance, into a multi-kelvin shift of the cold-trap depth at which water ice is stable on geological timescales. Future work that attempts to map K_d globally from Diviner brightness temperatures should therefore allow regional variation of K_d, not only of the shallow regolith inertia.

5 Conclusion

We have presented the first quantitative, per-site, single-parameter retrieval of the lunar deep regolith conductivity K_d from the restored Apollo HFE record, performed within the Hayne et al.

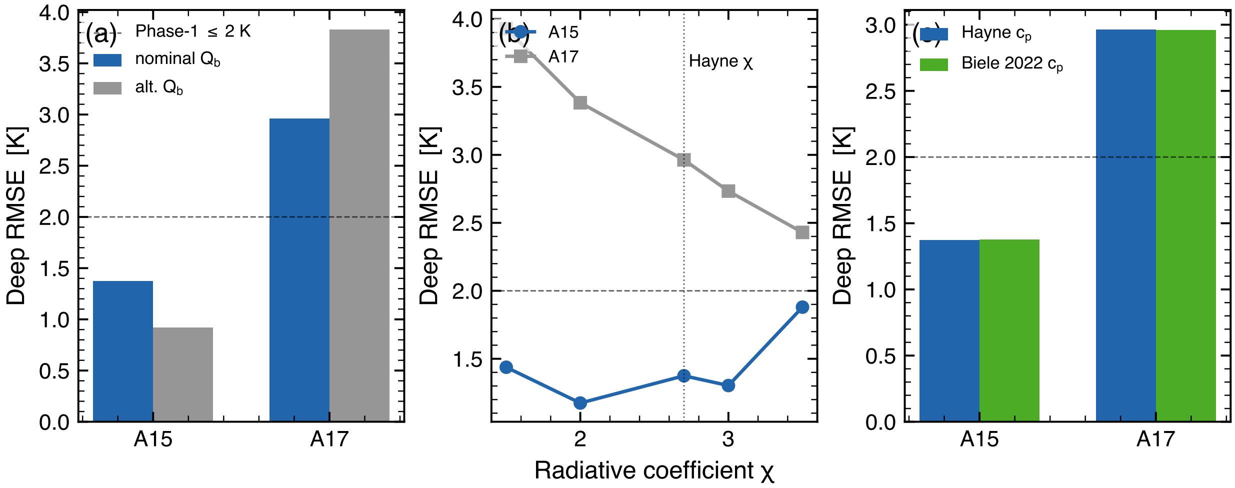

Figure 5. Parameter-sensitivity suite for the Hayne et al. (2017) reference model. Deep-sensor RMSE (with the dashed line at 2 K shown for reference) versus (a) basal heat flux Q_b, (b) radiative-conductivity coefficient χ (Hayne et al. (2017) default $\chi = 2.7$ marked), and (c) specific-heat parameterisation. Shaded ranges span the published uncertainty of each parameter; the deep RMSE shifts by < 0.5 K across all sweeps.

(2017) regolith functional form. The two landing sites require $K^*_{d,\mathrm{A15}} = 4.85 \pm 0.59$ and $K^*_{d,\mathrm{A17}} = 11.23 \pm 0.61$ mW/(m K) ($\sim 7.5\sigma$ contrast on the leave-one-out jackknife), and both reject the published global value 3.4 mW/(m K) at $\geq 2\sigma$. The K_d/Q_b degeneracy that conditions any single-borehole inversion leaves the per-site contrast invariant; the absolute A17 number is not similarly robust and should be quoted with that caveat. The borestem heat-short artefact above $z = 80$ cm is identified *a priori* from the diurnal-amplitude depth profile rather than from temperature residuals, yielding a hardware-based exclusion criterion that is identical at A15 and A17 and that isolates the deep-sensor RMSE used in the retrieval. Taken together, the result implies that lunar deep regolith conductivity is regionally heterogeneous and that the global parameter currently adopted in volatile-stability and instrument-design analyses should carry an uncertainty no smaller than the inter-site disagreement reported here.

Limitations

Three caveats should accompany any use of the per-site K_d retrieval reported here. *(i)* The absolute values are conditional on the adopted basal heat fluxes (§4); the per-site contrast is invariant under any global Q_b rescaling, but the absolute A17 value is not, and a $\sim$30% revision of $Q_{b,\mathrm{A17}}$ along the lines of the Saito et al. (2007); Nagihara et al. (2018) reanalyses brings $K^*_{d,\mathrm{A17}}$ into approximate agreement with the deep layer of the Martínez and Siegler (2021) 3-layer profile. *(ii)* The retrieval assumes that the Hayne et al. (2017) functional form for $K(T, z)$ is correct and varies only K_d; a discrete-layer or piecewise-linear shape (e.g. Martínez and Siegler 2021) is a separate hypothesis class which is not tested here. *(iii)* The diurnal-amplitude depth profile (Fig. 4) is used only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitisation noise floor.

Open Research

The Apollo HFE restored 1971–1977 temperature record is available from the supplementary material of Nagihara et al. (2018). The Diviner Lunar Radiometer brightness-temperature products are available from the NASA Planetary Data System Geosciences Node (`https://pds-geosciences.wustl.edu`). The 1-D Crank–Nicolson solver, the SPICE/DE440 surface-forcing chain, the K_d-sweep harness, the validation notebook, and the LaTeX sources for this manuscript and the supporting in-

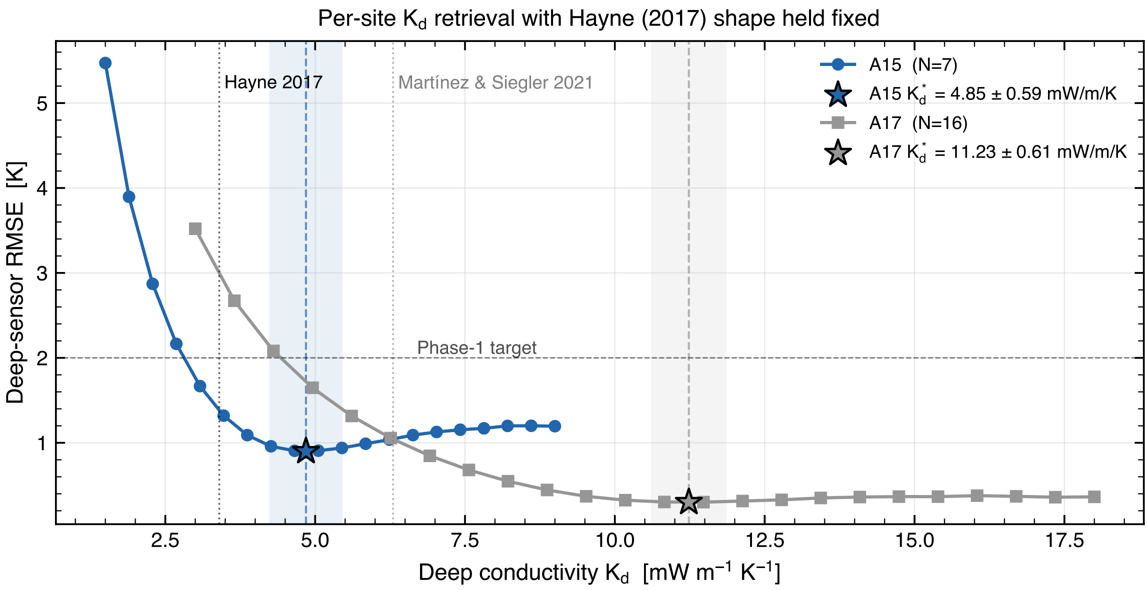

Figure 6. Per-site K_d retrieval under the Hayne et al. (2017) functional form. Deep-sensor RMSE (eq. 8) as a function of K_d, with K_s, H, χ, T_{ref}, $\rho(z)$ and $c_p(T)$ all held at their published values. Filled markers are the swept grid; the solid curve is the parabolic refinement through the three values bracketing the minimum; and the shaded band is the $\pm 1\sigma$ leave-one-out jackknife envelope. Vertical dashed lines mark the published Hayne et al. (2017) value ($3.4\,\mathrm{mW/(m\,K)}$) and the Martínez and Siegler (2021) 3-layer deep value ($6.3\,\mathrm{mW/(m\,K)}$). The two landing-site optima differ by $6.4 \pm 0.9\,\mathrm{mW/(m\,K)}$ ($\sim 7.5\sigma$).

formation are archived at `https://github.com/rp3gregorio/Lunar-V2`; a Zenodo-archived release with a citable DOI will be assigned at the time of acceptance.

Acknowledgements

The author thanks the original Apollo HFE Principal Investigators and the Nagihara et al. (2018) team for restoring the 1975–1977 portion of the record from the original tapes, without which this work would not have been possible.

References

Charles Acton, Nathaniel Bachman, Boris Semenov, and Edward Wright. A look toward the future in the handling of space science mission geometry. *Planetary and Space Science*, 150:9–12, 2018. doi: 10.1016/j.pss.2017.02.013. SPICE Toolkit; JPL DE440 ephemeris used here.

Charles H. Acton. Ancillary data services of NASA's Navigation and Ancillary Information Facility. *Planetary and Space Science*, 44(1):65–70, 1996. doi: 10.1016/0032-0633(95)00107-7.

J. L. Bandfield, P. O. Hayne, J.-P. Williams, B. T. Greenhagen, and D. A. Paige. Lunar surface roughness derived from LRO Diviner Radiometer observations. *Icarus*, 248:357–372, 2015. doi: 10.1016/j.icarus.2014.11.009.

J. Biele, E. Kuehrt, H. Senshu, N. Sakatani, K. Ogawa, M. Hamm, and M. Grott. Effects of dust layers on thermal emission from airless bodies. *Progress in Earth and Planetary Science*, 9:1–22, 2022. doi: 10.1186/s40645-022-00509-z.

P. O. Hayne, J. L. Bandfield, M. A. Siegler, A. R. Vasavada, R. R. Ghent, J.-P. Williams, B. T. Greenhagen, O. Aharonson, C. M. Elder, P. G. Lucey, and D. A. Paige. Global Regolith Thermophysical Properties of the Moon From the Diviner Lunar Radiometer Experiment. *Journal of Geophysical Research: Planets*, 122(12):2371–2400, 2017. doi: 10.1002/2017JE005387.

B. S. Hemingway, R. A. Robie, and W. H. Wilson. Specific heats of lunar soils, basalt, and breccias from the Apollo 14, 15, and 16 landing sites between 90 and 350 K. Open-File Report 73-149, U.S. Geological Survey, 1973. Polynomial fit subsequently adopted by Hayne et al. (2017) Appendix A.

G. Kopp and J. L. Lean. A new, lower value of total solar irradiance: Evidence and climate significance. *Geophys. Res. Lett.*, 38:L01706, 2011. doi: 10.1029/2010GL045777.

M. G. Langseth, S. J. Keihm, and K. Peters. Revised lunar heat-flow values. *Proc. Lunar Sci. Conf. 7*, pages 3143–3171, 1976.

A. Martínez and M. A. Siegler. A global thermal conductivity model for the surface of the Moon. *Journal of Geophysical Research: Planets*, 126(11):e2021JE006829, 2021. doi: 10.1029/2021JE006829.

S. Nagihara, W. S. Kiefer, P. T. Taylor, D. R. Williams, and Y. Nakamura. Examination of the long-term subsurface warming observed at the Apollo 15 and 17 sites utilizing the newly restored heat flow experiment data from 1975 to 1977. *Journal of Geophysical Research: Planets*, 123(5): 1125–1139, 2018. doi: 10.1029/2018JE005579.

D. A. Paige, M. C. Foote, B. T. Greenhagen, J. T. Schofield, S. Calcutt, A. R. Vasavada, et al. The lunar reconnaissance orbiter Diviner lunar radiometer experiment. *Space Science Reviews*, 150: 125–160, 2010. doi: 10.1007/s11214-009-9529-2.

W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery. *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press, 3rd edition, 2007.

Y. Saito, S. Tanaka, J. Takita, K. Horai, and A. Hagermann. Lunar heat flow experiment for SELENE-2. *Adv. Space Res.*, 40:1601–1607, 2007. doi: 10.1016/j.asr.2007.07.007.

A. R. Vasavada, J. L. Bandfield, B. T. Greenhagen, P. O. Hayne, M. A. Siegler, J.-P. Williams, and D. A. Paige. Lunar equatorial surface temperatures and regolith properties from the Diviner Lunar Radiometer Experiment. *Journal of Geophysical Research: Planets*, 117(E12), 2012. doi: 10.1029/2011JE003987.